Discrete-Time Fourier Transform Properties

Property	$x[n] \overset{DTFT}{\longleftrightarrow} X(e^{j\omega}) \quad y[n] \overset{DTFT}{\longleftrightarrow} Y(e^{j\omega})$
Linearity	$ax[n] + by[n] \overset{DTFT}{\longleftrightarrow} aX(e^{j\omega}) + bY(e^{j\omega})$
Time Shift	$x[n - n_o] \overset{DTFT}{\longleftrightarrow} e^{-j\omega n_o} X(e^{j\omega})$
Frequency Shift	$e^{j\gamma n} x[n] \overset{DTFT}{\longleftrightarrow} X(e^{j(\omega-\gamma)})$
Differentiation in Frequency	$-jnx[n] \overset{DTFT}{\longleftrightarrow} \frac{d}{d\omega} X(e^{j\omega})$
Convolution-Multiplication	$\sum_{k=-\infty}^{\infty} x[k]y[n-k] \overset{DTFT}{\longleftrightarrow} X(e^{j\omega})Y(e^{j\omega})$
Multiplication-Convolution	$x[n]y[n] \overset{DTFT}{\longleftrightarrow} \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\gamma})Y(e^{j(\omega-\gamma)})d\gamma$
Parseval's Theorem	$\sum_{n=-\infty}^{\infty} \lvert x[n]\rvert^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \lvert X(e^{j\omega})\rvert^2 d\omega$
Symmetry (for $x[n]$ real)	$X(e^{-j\omega}) = X^*(e^{j\omega})$ $\text{Real}\{X(e^{j\omega})\} = \text{Real}\{X(e^{-j\omega})\}$ $\text{Imag}\{X(e^{j\omega})\} = -\text{ Imag}\{X(e^{-j\omega})\}$ $\lvert X(e^{j\omega})\rvert = \lvert X(e^{-j\omega})\rvert$ $\arg\{X(e^{j\omega})\} = -\arg\{X(e^{-j\omega})\}$